JECRC University, Jaipur
Department of Physics
B. Tech Ist Year – Ist Semester July-Dec 2025-26
Applied Physics (DPH001B)
Assignment-I

Submission Date: 15-9-2025

A. Very Short Answer Questions (each question carries 2 marks)

1. Explain, why the Compton Effect is not observed experimentally for visible rays?
2. Define the concept of operator in quantum mechanics. Write the Hamiltonian, Kinetic energy and angular momentum operators?
3. A nucleon is confined to a nucleus of diameter $5\times10^{-4}m$. Calculate the minimum uncertainty in the momentum and kinetic energy of the nucleon.
4. Why the concept of group velocity is needed?
5. State the difference between degeneracy and non-degeneracy.

B. **Answer the following briefly (each question carries 7 marks)**

1. (a) In Compton scattering, the incident photon have wavelengths $3\times10^{-10}\,m$. Calculate the wavelength of scattered radiation if they are viewed at the angle of 60^0 to the direction of incidence.
   (b) Calculate the kinetic energy imparted to the recoil electron, when X-ray of wavelength $1A^0$ are scattered from a carbon block at right angle.
2. For $n_x + n_y + n_z = 6$, consider an electron is trapped in a rectangular box with sides a, b, c. calculate the energy eigen values and eigenwave function for the followings:
   (a) if $a = b \neq c$; (b) if a≠b≠ c, (c) a=b=c; and study the effect of degeneracy occurs in all cases.
3. What do you understand with Quantum mechanical tunneling? On account of this explain the phenomena of Alpha decay.
4. The wave function of a particle in its ground state in one dimensional box of length L is given by $\psi = \sqrt{2/L}\sin(n\pi/L)x$. Calculate the probability of finding the particle can be found between x-0 and x=L/n, when it is in nth state.

C. **Answer the followings explicitly (each question carries 11 marks)**

1. Using a schematic diagram explain the change in wavelength in Compton Effect. Derive relations for change in energy of incoming photon and what is maximum kinetic energy gained by recoil electron?
2. A particle is trapped in confined to move in one dimensional in a rigid box of width ‘a’. Use time independent Schrödinger’s equation, carry out the expression for energy eigen values and eigen wave function. Also plot wavefunction and probabilities for first three energy levels.
3. Define the term “Wave Packet” and conditions for its validation. Give the reason that why the wave function ψ itself has no physical significance? State clearly the conditions of normalizability and orthogonality.